



Research paper

SwinIBSR: Towards real-world infrared image super-resolution

Yifeng Shi, Nan Chen^{*}, Yuesheng Pu, Jiqing Zhang, Libin Yao

Kunming Institute of Physics, Kunming, Yunnan Province, China

ARTICLE INFO

Keywords:

Super-resolution
Degradation model
Real-world
Deep learning
Infrared image

ABSTRACT

Traditional infrared image super-resolution (SR) methods often fall short in real-world scenarios, particularly when dealing with images degraded by various forms of noise and artifacts. Addressing this gap, this paper presents SwinIBSR, a blind SR approach optimized for real-world infrared imaging conditions. SwinIBSR introduces a practical degradation model, specifically designed to simulate the complex degradation processes encountered in real-world infrared imaging. Additionally, a mixed training strategy using both infrared and visible light datasets greatly enhances the network's generalization capabilities, leading to the production of high-quality SR images. Building upon the Transformer-based architecture of SwinIR, SwinIBSR is adept at learning intricate mappings from low-resolution to high-resolution images. Our comprehensive experiments validate the effectiveness of SwinIBSR, showcasing its outstanding performance in processing real-world infrared images and its notable superiority over existing methods regarding visual quality. This work contributes to the field of infrared image SR, offering insights in enhancing SR techniques for practical applications.

1. Introduction

Infrared detectors have been widely used in surveillance, temperature measurement, fault detection, remote sensing, etc. Due to the low frequency of the infrared spectrum and the small scale of the infrared focal plane array, the imaging quality of infrared images is not as good as that of the visible spectrum. To improve the signal-to-noise ratio (SNR), the infrared detector is generally designed with a large pixel size, which can enhance the sensitivity of a single pixel, but this will sacrifice the spatial resolution. However, designing a high-resolution infrared sensor needs to reduce the pixel size and increase the sensor size, which is expensive and difficult to achieve. As a result, most infrared detectors maintain a high SNR at the expense of resolution. In order to obtain high-resolution and high-SNR infrared images, researchers use software-driven algorithms called image super-resolution (SR) to reconstruct high-resolution (HR) images from low-resolution (LR) infrared images.

Deep learning has achieved great success in visible image SR in recent years [1,2]. The main reason is that deep learning can automatically extract features from images and learn the mapping from LR to HR images. In the field of infrared image SR, many researchers have also proposed approaches based on deep learning [3–9]. However, there is still a significant gap in the visual quality of real-world applications. Specifically, (1) The existing methods are mainly based on non-blind image SR, so they are not practical for real-world applications. (2) Due to the lack of training samples (i.e., the HR infrared images),

these methods often overfit the training data and cannot generalize well to the test data. (3) These methods emphasize objective metrics such as peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) [10], ignoring the human eye's subjective perception. Moreover, some studies [11–13] indicate that PSNR and SSIM correlate poorly with human perception of visual quality. Therefore, the visual quality of these methods is not realistic, and the artifacts are apparent. (4) The existing methods are almost based on CNN, which is unsuitable for long-range dependencies between pixels.

To address the above problems, we introduce SwinIBSR, a blind SR method for infrared images, which builds upon the SwinIR architecture. The main contributions of this paper are summarized as follows:

- (1) We propose a practical degradation model for infrared SR. This model effectively simulates the complex degradation processes prevalent in real-world infrared imaging, ensuring that the generated LR images closely resemble natural scenes.
- (2) Leveraging our degradation model, we generate a vast array of synthetic training samples. This strategy significantly enhances the network's ability to generalize to real-world data. Additionally, we introduce a mixed training methodology using both infrared and visible image datasets, leading to improvements in crucial visual aspects such as texture and edge details.
- (3) We adopt a generative adversarial network (GAN) framework, utilizing a U-Net discriminator with spectral normalization, to

^{*} Corresponding author.

E-mail address: chennan_kip@163.com (N. Chen).

produce SR images that closely simulate reality. Our training process combines L1 loss, perceptual loss, adversarial loss, and LDL loss, effectively reducing artifacts and elevating the visual quality of reconstructed images.

- (4) By employing SwinIR as the generator and incorporating advancements from Swin Transformer V2, our method capably learns long-range pixel dependencies. This leads to superior performance compared to existing methods. The Transformer's proven efficacy in visible image SR is further extended to infrared image SR, showcasing its adaptability.

2. Related work

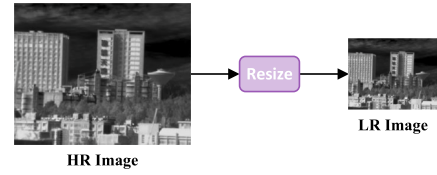
Image super-resolution. Image super-resolution, also known as image super-sampling, is a kind of low-level vision task that aims to reconstruct a HR image from its LR counterpart. It is an ill-posed problem since multiple possible solutions exist for a given LR image. For convenience, we omit the image channel and define the degraded LR image as $I_{LR} \in \mathbb{R}^{h \times w}$, the reconstructed SR image as $I_{SR} \in \mathbb{R}^{H \times W}$, and the ground-truth HR image as $I_{HR} \in \mathbb{R}^{H \times W}$. The purpose of the SR task is to find a mapping function f that can reconstruct the SR image from the LR image, i.e., $I_{SR} = f(I_{LR})$. In deep learning-based SR methods, the mapping function f can be represented as a deep neural network (DNN) with parameters θ . The goal can be formulated as the following optimization problem:

$$\theta^* = \arg \min_{\theta} \mathcal{L}(I_{HR}, f(I_{LR}; \theta)) \quad (1)$$

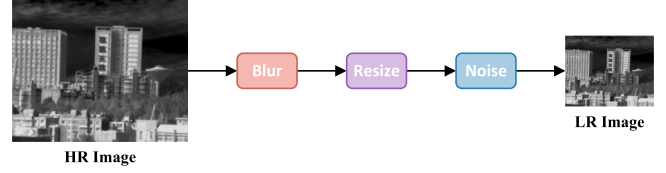
where $\mathcal{L}$ is the loss function, the loss function is usually defined as the pixel-wise difference between the reconstructed SR image and the ground-truth HR image. The optimization problem in Eq. (1) is solved by minimizing the loss function concerning the network parameters θ .

Inspired by the first deep learning-based SR approach SRCNN [14], many researchers have proposed various approaches. FSRCNN [15] proposed a deeper network structure and used deconvolution layers to promote performance and reduce computational costs. After the concept of residual structure [16] was proposed, the depth of the network was further increased [17–21]. Additionally, researchers have proposed a multi-scale structure to extract features of various receptive fields [22,23]. To achieve more realistic SR images, GAN [24] was introduced into the SR task as a penalty item [25–29]. However, the GAN-based methods are challenging to train and usually generate artifacts. In order to solve these limitations, researchers have proposed a variety of loss functions, such as the perceptual loss [30], and LDL loss [31].

Recent years have seen the rapid development of deep learning-based infrared image SR methods. Sun et al. [3] introduced the zoom mechanism (i.e., a combination of deconvolution layer and pooling layer) to enlarge and shrink the feature map, which can effectively perform the nonlinear mapping. Suryanarayana et al. [4] proposed a multi-scale saliency detection to extract the small and large salient feature maps of the LR image and then used a residual network to reconstruct the SR image. The processing is performed in the wavelet domain. Yao et al. [5] combined a discriminative dictionary and deep residual network to achieve more accurate results. Oz et al. [6] employed a depthwise separable convolution network to accelerate the inference process. Zou et al. [7] adopted a U-Net structure to fuse the multi-scale features. Zhang et al. [8] proposed a multi-image SR approach composed of a neighboring frames alignment module, a spatial-temporal feature fusion module, and a reconstruction module. Huang et al. [9] proposed a lightweight heterogeneous CNN, and adversarial training was also adopted.



(a) Simple Downsampling (Eq.(3))



(b) Downsampling with Blur and Noise (Eq.(4))

Fig. 1. Degradation models used in previous studies.

Degradation model. In infrared SR, the LR image I_{LR} is derived from the HR image I_{HR} through a complex degradation process. This process encompasses blurring, downsampling, noise, and other unknown factors, and can be mathematically represented as:

$$I_{LR} = D(I_{HR}; \theta_D) \quad (2)$$

where D denotes the degradation map $\mathbb{R}^{H \times W} \rightarrow \mathbb{R}^{h \times w}$, and θ_D represents its parameters. The SR task thus involves modeling the inverse map D^{-1} , i.e., $I_{SR} = D^{-1}(I_{LR})$.

Previous infrared SR studies [3,4,6–9] have commonly employed two relatively simple degradation models, as depicted in Fig. 1:

$$D(I_{HR}; \theta_D) = (I_{HR}) \downarrow_s \quad (3)$$

$$D(I_{HR}; \theta_D) = (I_{HR} \otimes \kappa) \downarrow_s + n \quad (4)$$

Here, $\downarrow_s$ indicates downsampling with scale factor s , $I_{HR} \otimes \kappa$ signifies blurring with a Gaussian kernel κ , and n is additive white Gaussian noise.

However, real-world infrared imaging scenarios are far more complex, involving various external factors like lighting conditions, sensor noise, and environmental influences. For example, blurring might occur due to non-Gaussian kernels, noise might not be Gaussian, and downsampling could involve more complex mechanisms. Thus, the simple degradation models in Eqs. (3) and (4) fall short in accurately simulating the multifaceted real-world degradation process, making them unsuitable for practical applications.

Blind super-resolution. Blind image SR [32] refers to reconstructing an HR image from the counterpart one with unknown and complex degradation models. It is generally acknowledged that SR methods' performance will degrade and be accompanied by unpleasing artifacts if the assumption degradation model deviates from the real-world situation. Therefore, the degradation model is crucial for applying SR methods in real scenarios. In the field of visible SR, there has been much excellent work [33–36] on blind image SR to address this problem.

Transformer. Recently, Transformer (also known as attention mechanism) [37] has been widely used in computer vision tasks [38–44]. Transformer is a robust network structure that can learn the long-range dependencies between pixels, which is very suitable for SR tasks. Previous works [41,44–47] have shown that Transformer can achieve better performance than CNN in the visible image SR field.

3. Method

3.1. Overview

This section introduces the overarching structure and operational flow of our proposed SR method, as shown in Fig. 2.

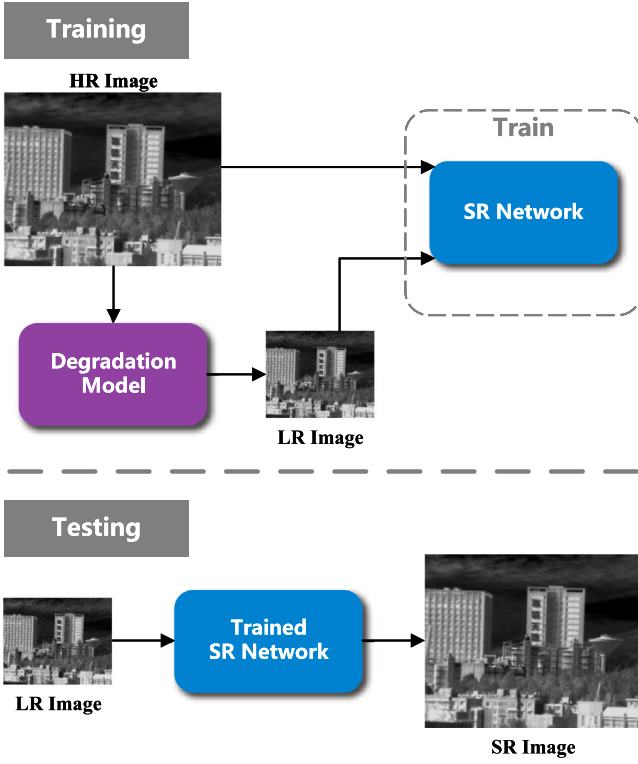


Fig. 2. Overview of the proposed SR method's training and testing phases. The left side depicts the training process, where HR images are degraded to create LR counterparts for network training. The right side illustrates the testing process, where the trained network enhances LR images to generate SR outputs.

Training phase. The upper part of Fig. 2 illustrates the training schematic, where our degradation model plays a pivotal role. It processes HR images to generate their LR counterparts, forming HR-LR image pairs. These pairs are then utilized to train the SR network, optimizing it to accurately reconstruct HR images from LR inputs.

Testing phase. The lower part of Fig. 2 details the testing phase, showcasing how the trained SR network applies to unseen LR images. In this phase, the input LR images are transformed by the network to produce SR images.

3.2. Degradation model

Given the intrinsic relationship between the super-resolution (SR) problem and its degradation model, where they serve as inverse processes to each other, the design of the degradation model is crucial and can significantly affect the performance of the SR method. Drawing inspiration from the degradation models presented in [35,36], and with a specific focus on overcoming the distinct challenges posed by infrared imaging, we introduce a novel unified gated second-order degradation model (see Fig. 3). This model is meticulously crafted to mirror the complex degradation processes encountered in real-world infrared imaging scenarios, encompassing not only a range of common degradation types but also the simulation of column striping artifacts that are characteristic of infrared detectors.

The mathematical formulation of our unified gated second-order degradation model is represented as follows:

$$I_{LR} = D^2(I_{HR}) = (D \circ D)(I_{HR}) \quad (5)$$

$$D = \sigma_g(D_1) \circ \sigma_g(D_2) \circ \dots \circ \sigma_g(D_m) \quad (6)$$

where D^2 denotes the second-order composite degradation map, systematically applying a sequence of multiple degradation units to the

HR image I_{HR} . Each degradation unit, denoted by D_i , is selected from a predefined set that includes blurring (D_b), application of a sinc filter (D_s), resizing (D_r), addition of noise (D_n), introduction of column striping (D_c), and JPEG compression artifacts (D_j). These units are controlled by a gated activation function σ_g , defined as

$$\sigma_g(D_i) = \begin{cases} D_i, & g = 1, \\ I, & g = 0, \end{cases} \quad (7)$$

with I denoting the identity operation, and g being a binary gate that controls the activation of each degradation unit based on a predefined probability p . In the following, we will introduce the degradation units in detail.

Blur. Blur is a common degradation in the real world. We use convolution to simulate the blur effect. For different training samples, the convolution kernel k is randomly chosen from the Gaussian blur kernel, generalized Gaussian blur kernel [48], and plateau-shaped blur kernel. Given a kernel size $2t + 1$ and a coordinate $(i, j) \in [-t, t]^2$, their probability density functions are as follows:

$$k(i, j) = \frac{1}{N} \exp\left(-\frac{1}{2}x^T \Sigma^{-1}x\right) \quad (8)$$

$$k(i, j) = \frac{1}{N} \exp\left(-\frac{1}{2}x^T \Sigma^{-1}x\right)^\beta \quad (9)$$

$$k(i, j) = \frac{1}{N} \frac{1}{1 + (x^T \Sigma^{-1}x)^\beta} \quad (10)$$

where N is the normalization factor, Σ is the covariance matrix, β is the shape parameter, and $x = [i, j]^T$ is the vector of the coordinates of the pixel. The covariance matrix Σ is further represented as

$$\begin{aligned} \Sigma &= R \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} R^T \\ &= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \end{aligned} \quad (11)$$

where R is the rotation matrix, σ_1 and σ_2 are the standard deviations of the Gaussian distribution along the two axes, and θ is the rotation angle. σ_1 , σ_2 , and θ are also randomly sampled from the uniform distribution for different training samples. By randomly sampling the parameters of the blur kernel, we can simulate the blur effect of various strengths and directions.

Sinc filter. Sinc filter is a type of blur often used to simulate ringing artifacts. Ringing artifacts appear as spurious signals near sharp transitions in a signal. In the image, they appear as bands or “ghosts” near the edges. The sinc filter kernel is defined as

$$k(i, j) = \frac{1}{N} \frac{\omega_c}{2\pi\sqrt{i^2 + j^2}} J_1\left(\omega_c \sqrt{i^2 + j^2}\right) \quad (12)$$

$$J_1(x) = \frac{1}{\pi} \int_0^\pi \cos(\tau - x \sin \tau) d\tau \quad (13)$$

where N is the normalization factor, ω_c is the cutoff frequency, and J_1 is the first-order Bessel function of the first kind. The cutoff frequency ω_c is randomly sampled from the uniform distribution for different training samples.

Resize. Downsampling is a direct way to reduce the resolution of the image. Since the degradation process is second-order, we can use downsampling and upsampling to cover more cases (i.e., resize operation). Thus we can zoom in or zoom out the image at first resize operation and then resize the image to the target size at the second resize operation. The interpolation method is randomly chosen from linear, bicubic, and area interpolation.

Noise. To perform SR in the real world, we have to consider the effect of noise. We use additive Gaussian noise and Poisson noise to simulate the noise effect. Since the infrared image is a single channel, we only consider the grayscale noise. For brevity, the noise of different pixels

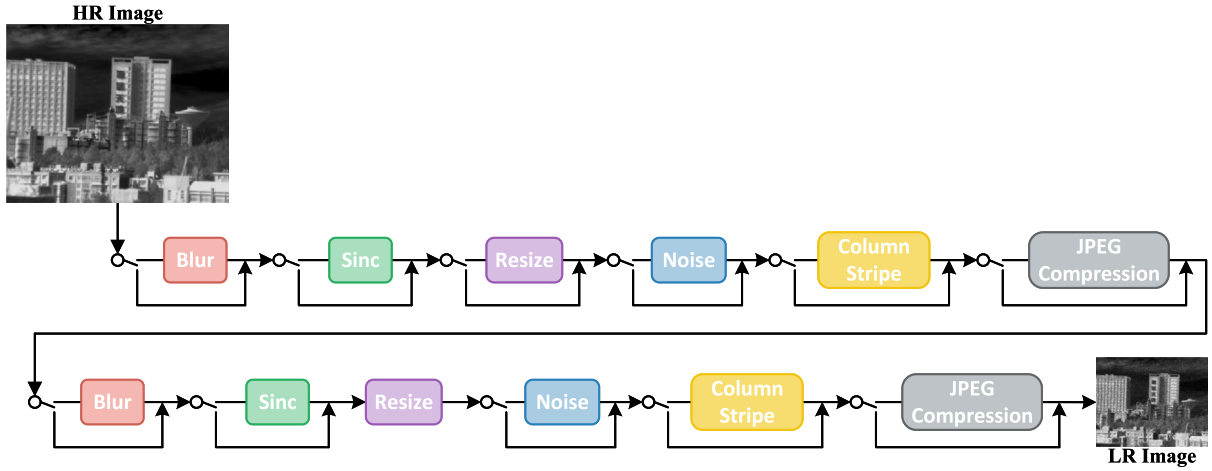


Fig. 3. Overview of the unified gated second-order degradation model.

is independent and identically distributed (i.i.d.). The Gaussian noise is sampled from the normal distribution with zero mean and a standard deviation σ . The Poisson noise is sampled from the Poisson distribution with a mean λ , then shifts the noise's mean to zero. σ and λ are randomly chosen from the uniform distribution. In addition, noise type is also randomly chosen for each training sample.

Column stripe. Column striping, a prevalent issue in infrared imaging, arises primarily due to the use of shared column structures in the readout circuitry of the detector. This architectural design leads to non-uniform responses across different columns of the detector elements, manifesting as visually distinct vertical stripes in the captured images. To accurately simulate this effect, we draw inspiration from the approach presented in DLS-NUC [49]. In this model, the column striping effect is represented as a polynomial function of the pixel values or coordinates within each column. The mathematical representation of this simulation is formulated as

$$Y(i, j) = \sum_{p=0}^3 a_p(j) X(i, j)^p \quad (14)$$

where $Y(i, j)$ signifies the intensity of the column striping effect at each pixel in row i and column j . Diverging from the uniform distribution method employed in DLS-NUC, we utilize a normal distribution for the polynomial coefficients $a_p(j)$. This approach yields more naturally varied striping patterns, closely mirroring the stochastic non-uniformities typical in infrared detector arrays. $X(i, j)$ denotes the original pixel values, forming the basis of this simulation.

JPEG compression. JPEG compression [50] is a commonly used image compression standard for digital images. Many infrared images are stored in JPEG format to reduce transmission and storage pressure. However, JPEG compression is a lossy method, which will cause quality loss and artifacts. Therefore, we use JPEG compression to simulate real-world scenarios, which enables our network to remove artifacts. The quality factor $q \in [0, 100]$ controls the image quality, where a higher q means better image quality. We randomly choose q from the uniform distribution for each training sample.

3.3. Network architecture

Inspired by SwinIR [44] and Swin Transformer V2 [42,43], we propose SwinIBSR to perform infrared image blind SR. The generator architecture is shown in Fig. 4 and comprises three parts: shallow feature extraction, deep feature extraction, and reconstruction. The shallow feature extraction and reconstruction modules are the same as SwinIR. The details of each module are described below.

Shallow feature extraction. The shallow feature extraction module, consisting of a single 3×3 convolutional layer H_{SF} with 96 channels, is strategically chosen for its efficiency and effectiveness in information aggregation at the network's initial stage. This can be formulated as

$$F_0 = H_{SF}(I_{LR}) \quad (15)$$

where $I_{LR} \in \mathbb{R}^{c \times h \times w}$ denotes the LR image, and $F_0 \in \mathbb{R}^{d \times h \times w}$ represents the shallow feature map. In the context of Transformer-based SR tasks, employing a convolutional layer at the outset facilitates efficient information gathering, which is crucial since direct application of Transformer layers to image patches can result in undesirable blocky artifacts in the reconstructed SR image. This approach also aligns with the linear projection and patch embeddings strategy utilized in conventional vision Transformer [38,42,43]. Recent findings from [51] corroborate our approach, suggesting that early-stage convolutions can significantly improve optimization stability and model performance by injecting a minimal convolutional inductive bias into the ViT models.

Deep feature extraction. The deep feature extraction module integrates 4 Residual Swin Transformer Blocks (RSTB), each comprising 6 Swin Transformer Layers (STL). Each STL layer, key to feature extraction, employs a shifted window-based multi-head self-attention mechanism, a multi-layer perceptron (MLP), and layer normalization. A convolutional layer, positioned at the end of this module and within each RSTB, aggregates information effectively. The processing flow of the deep feature extraction module is formalized as

$$F_i = H_{RSTB_i}(F_{i-1}), \quad i \in \{1, 2, 3, 4\} \quad (16)$$

$$F_{DF} = H_{CONV}(F_4) \quad (17)$$

where $F_{DF} \in \mathbb{R}^{d \times h \times w}$ represents the deep feature map, which is subsequently input into the reconstruction module.

While the core structure of this module derives from SwinIR, we have incorporated advancements from Swin Transformer V2, particularly to augment blind SR task performance. These improvements include: (1) Utilizing residual-post-normalization instead of pre-normalization in STL, which allows for more stable and effective feature learning. (2) Adopting scaled cosine attention and log-spaced continuous relative position bias within the self-attention mechanism of STL. This supersedes the traditional dot-product attention and relative position bias, thereby improving the model's capacity for capturing finer details and spatial relationships in images.

This design facilitates the efficient capture of long-range dependencies and context-aware feature processing, vital for SR tasks. Specifically, the shifted window approach within the STL significantly enhances the model's capability to aggregate information across different

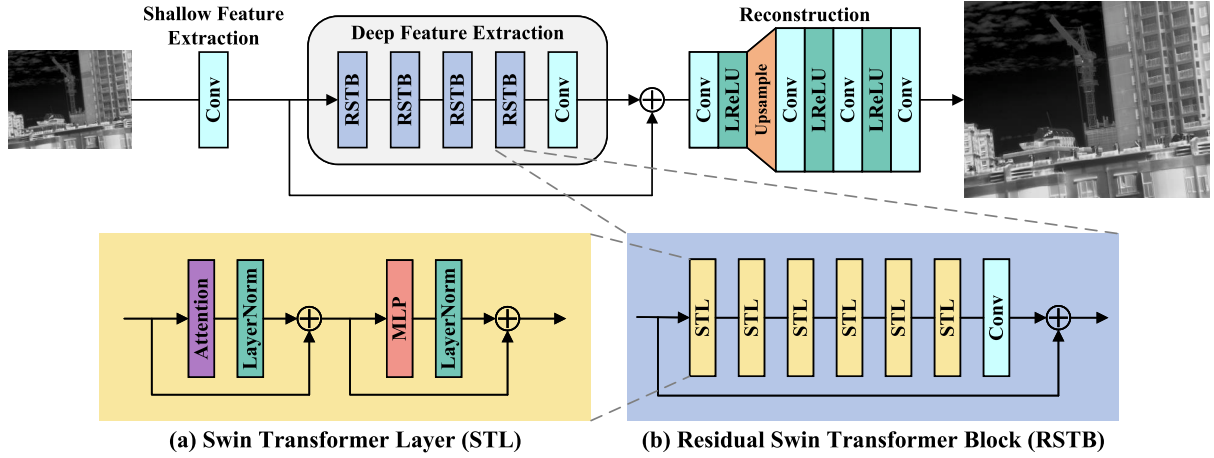


Fig. 4. The network architecture of SwinIBSR.

regions of the image. This aspect is essential for addressing the complex degradation processes typical in real-world scenarios, where images often undergo various forms of blurring, noise, and downsampling. The sequential layering of STLs within each RSTB methodically refines the feature maps, adapting to and compensating for these real-world degradation effects. This module effectively contributes to the reconstruction of detailed HR images from their LR counterparts, aligning with the practical demands of infrared imaging in real-world applications.

Reconstruction. The reconstruction module in our network, intended for $2\times$ SR, is depicted in Fig. 4, showcasing the arrangement of convolutional layers, leaky ReLU activations, and an upsampling layer. The process of the reconstruction module can be formulated as

$$I_{SR} = H_{REC}(F_0 + F_{DF}) \quad (18)$$

where $I_{SR} \in \mathbb{R}^{c \times H \times W}$ is the reconstructed SR image. The residual connection between the shallow and deep feature maps facilitates the fusion of low and high-frequency information. The upsampling operation is implemented via nearest neighbor interpolation, a method not only effective in reducing the blocky effect commonly associated with upscaling but also advantageous due to its high computational efficiency.

Adversarial network. In the SwinIBSR network, we adopt the discriminator architecture from Real-ESRGAN [35] to train our generative adversarial network. This discriminator utilizes a U-Net design with skip connections. Additionally, it incorporates spectral normalization (SN), which is a critical feature for stabilizing the training process. The U-Net discriminator, enhanced by SN, is particularly adept at providing detailed, per-pixel feedback to the generator. This specific capability is crucial for enhancing local textures and achieving realistic details in the SR images, particularly important for infrared image blind SR tasks.

3.4. Loss function

The loss function plays a vital role in optimizing. However, previous infrared SR methods usually use L1 loss, L2 loss, and Charbonnier loss [52], which can get high PSNR or SSIM but have poor visual quality. In this paper, we use a combination of L1 loss, perceptual loss [30], LDL loss [31], and adversarial loss [26,28] to train the network. The loss function is formulated as

$$\mathcal{L}_G = \lambda_1 \mathcal{L}_1 + \lambda_p \mathcal{L}_p + \lambda_{LDL} \mathcal{L}_{LDL} + \lambda_{G_a} \mathcal{L}_{G_a} \quad (19)$$

$$\mathcal{L}_D = \lambda_{D_a} \mathcal{L}_{D_a} \quad (20)$$

where $\mathcal{L}_G$ and $\mathcal{L}_D$ are the loss function of the generator and discriminator, respectively. $\mathcal{L}_1$ is the L1 loss, $\mathcal{L}_p$ is the perceptual loss, $\mathcal{L}_{LDL}$

is the LDL loss, $\mathcal{L}_{G_a}$ is the adversarial loss of the generator, and $\mathcal{L}_{D_a}$ is the adversarial loss of the discriminator. λ_1 , λ_p , λ_{LDL} , λ_{G_a} , λ_{D_a} are the trade-off parameters of the loss function. The details of each loss function are described below.

L1 loss. The L1 loss is defined as

$$\mathcal{L}_1 = \|I_{HR} - I_{SR}\|_1 \quad (21)$$

The L1 loss measures the pixel-wise difference between the reconstructed SR and HR images. The L1 loss is widely used in the SR task and can get high PSNR and SSIM.

Perceptual loss. The perceptual loss is defined as

$$\mathcal{L}_p = \sum_l \|\phi_l(I_{HR}) - \phi_l(I_{SR})\|_1 \quad (22)$$

where ϕ_l is the feature map of the l th layer of the pre-trained VGG-19 network [53]. Since deep features represent semantic information, the perceptual loss focuses more on the image's visual quality instead of forcing them to match pixels exactly.

LDL loss. The LDL loss is defined as

$$\mathcal{L}_{LDL} = \|M_{refine} \cdot (I_{HR} - I_{SR})\|_1 \quad (23)$$

where $M_{refine} \in \mathbb{R}^{1 \times H \times W}$ is the mask of artifacts regions (see [31] for details). As known, GAN-based methods usually generate artifacts in the SR image. The LDL loss can effectively suppress the artifacts in the SR image. Its core idea is to calculate the artifact regions in the SR image and then penalize the regions in the loss function.

Adversarial loss. Due to the space limitation, we use I and $\hat{I}$ to represent the HR and SR images, respectively. The adversarial loss is defined as

$$\mathcal{R}_{I,I} = \sigma(D(I) - \mathbb{E}_I[D(\hat{I})]) \quad (24)$$

$$\mathcal{R}_{\hat{I},I} = \sigma(D(\hat{I}) - \mathbb{E}_I[D(I)]) \quad (25)$$

$$\mathcal{L}_G = -\mathbb{E}_I[\log(1 - \mathcal{R}_{I,I})] - \mathbb{E}_{\hat{I}}[\log \mathcal{R}_{\hat{I},I}] \quad (26)$$

$$\mathcal{L}_D = -\mathbb{E}_{\hat{I}}[\log \mathcal{R}_{\hat{I},I}] - \mathbb{E}_I[\log(1 - \mathcal{R}_{I,I})] \quad (27)$$

where D is the discriminator network, $\mathbb{E}$ is the expectation operator, and σ is the sigmoid function. $\mathcal{R}_{I,I}$ represents the probability that the HR image is more realistic than the SR image, and $\mathcal{R}_{\hat{I},I}$ represents the probability that the SR image is more realistic than the HR image. Driven by the adversarial loss, the discriminator learns to distinguish the HR and SR images, and the generator learns to generate more realistic SR images to fool the discriminator.

4. Experiments

4.1. Implementation

Datasets. DNN-based methods need a large amount of training data to achieve good performance. However, the infrared SR task is still in its infancy, and there are few datasets for training. Some previous works used a small amount of infrared data for training, which led to overfitting and poor generalization performance. Other works pre-trained the network on the visible datasets and then fine-tuned the network on the infrared datasets. However, the fine-tuning strategy can only accelerate the convergence of the initial stage of training and cannot solve the problem of lack of data.

In this paper, we utilize publicly available infrared datasets (from InfiRay Infrared Open-Source Platform) and visible datasets (DIV2K [54], and Flickr2K [55]) for mixed training of the network. The infrared datasets comprise 2079 images with a resolution of 1280×1024 and 6980 images with a resolution of 640×512 . The visible datasets contain a total of 3550 images of varying resolutions. For the visible images, we first convert them to the YCbCr color space and then extract the luminance channel. All these images are high-quality HR images.

To accommodate batch training, we crop the HR images into 224×224 patches, and the number of HR image patches is 307360. We degrade the HR images to get the LR-HR image pair with the method described in Section 3.2. Note that each training sample's degradation process and parameters are randomly selected. We also perform data augmentation on the training data. Specifically, we randomly flip the LR-HR image pair horizontally and vertically and then randomly rotate the LR-HR image pair by 90 degrees. Before feeding the LR-HR image pair into the network, we normalized the LR image and HR image to have a mean of 0 and a variance of 1 over the entire dataset. During the training process, we use sharpened HR image and SR image to calculate the L1 loss and perceptual loss to visually improve the sharpness of the SR image [35].

Degradation details. In order to synthesize as many samples as possible, our unified gated second-order degradation model takes multiple random parameters. Following the setting of Real-ESRGAN [35], the degradation parameters are set as follows.

The probability p for every degradation unit is set to 0.5. The kernel size of blur and sinc filter is randomly chosen from the $\{3, 5, 7, \dots, 21\}$. Gaussian blur kernel, generalized Gaussian blur kernel, and plateau-shaped blur kernel are adopted with a probability of $\{0.7, 0.15, 0.15\}$. Blur standard deviations σ_1 and σ_2 are sampled from $[0.2, 3]$. Blur rotation angle θ is sampled from $[0, \pi]$. The shape parameter β is sampled from $[0.5, 4]$ and $[1, 2]$ for generalized Gaussian and plateau-shaped blur kernels, respectively. Sinc filter cutoff frequency ω_c is sampled from $[\pi/3, \pi]$. Gaussian and Poisson noise are added with a $\{0.5, 0.5\}$ probability. Gaussian noise standard deviation σ is sampled from $[1, 30]$ ($[1, 25]$ for the second degradation process), and Poisson noise parameter λ is sampled from $[0.05, 3]$ ($[0.05, 2.5]$ for the second degradation process). For each polynomial term p , the coefficients $a_p(j)$ are independently sampled from the normal distribution $\mathcal{N}(0, 0.05)$ for every column j , contributing to the column striping simulation. The quality factor q of JPEG compression is sampled from $[30, 95]$.

Training details. We adopt AdamW optimizer [56] and Adam optimizer [57] to train the generator (SwinIBSR) and discriminator (i.e., U-Net discriminator with spectral normalization), respectively. $\beta_1 = 0.9$, $\beta_2 = 0.999$ are used for both optimizers. We adopt a cosine annealing learning rate scheduler [58] to decay the learning rate. The batch size is set to 16. We also adopt an exponential moving average (EMA) approach to update the generator parameters for more stable training. The EMA decay rate is set to 0.999. The training process is divided into two stages. In the first stage, the generator is trained with the L1 loss. The learning rate is set to 2×10^{-4} and is decreased from 2×10^{-4} to 1×10^{-6} in 45 epochs. In the second stage, the generator

and discriminator are trained using the loss function in Eqs. (19) and (20). The trade-off parameters λ_1 , λ_p , λ_{LDL} , λ_{G_a} , λ_{D_a} are set to 1, 1, 1, 0.1, 1, respectively. The learning rate is set to 1×10^{-4} and is decreased from 1×10^{-4} to 5×10^{-7} in 50 epochs. The model is implemented using PyTorch framework and trained on a single NVIDIA V100 GPU.

4.2. Ablation study

In this section, we conduct an extensive ablation study to evaluate the impact of various components and configurations on the performance of our proposed SwinIBSR model.

Variant configurations. Table 1 outlines the distinct configurations of each SwinIBSR variant. These configurations are crafted to examine the individual and combined effects of various model components, including features like residual-post-normalization and scaled-cosine attention, as well as a series of loss functions such as $\mathcal{L}_1$, $\mathcal{L}_p$, $\mathcal{L}_{G_a}$, and $\mathcal{L}_{LDL}$. The study also explores how different training datasets, encompassing both infrared and visible light datasets, impact the model's performance, along with the influence of the proposed degradation model.

Experimental settings. The effectiveness of each SwinIBSR variant is thoroughly assessed through both quantitative and qualitative means. Our study incorporates widely used visible light SR datasets, including Set5 [59], Set14 [60], BSD100 [61], and Urban100 [62], alongside infrared test datasets.

While many SR researches traditionally rely on PSNR and SSIM for evaluating the objective performance of SR methods, these metrics primarily reflect the similarity in global or local image mean and variance. Despite yielding high PSNR and SSIM scores, images rich in high-frequency details may still lack visual appeal. Studies have suggested that PSNR and SSIM may not always align with perceived visual quality in SR images [11–13]. Therefore, we have also included the natural image quality evaluator (NIQE) [63] score, a perceptual quality metric based on image naturalness, which does not rely on any distortion or subjective human opinions. A lower NIQE score signifies better visual quality. This metric choice aligns with our focus on perceptual quality over pixel-level accuracy, especially critical for real-world infrared imaging applications.

The outcomes are quantitatively depicted in Table 2, which includes PSNR and SSIM metrics for visible light images, and NIQE scores for both visible and infrared images. As ground truth data is unavailable for infrared images, we exclusively use the NIQE score for their evaluation. To avoid boundary artifacts, we cropped the borders of the images by 5 pixels. For a more intuitive understanding of the performance impact, Fig. 5 visually contrasts the results of each variant.

Discussion on network structure. The results from SwinIBSR-V1 to SwinIBSR-V3 reveal progressive improvements in PSNR, SSIM and NIQE, indicating enhanced structural fidelity and improved perceptual quality. These enhancements are visually evidenced in Fig. 5, which shows a noticeable increase in detail and clarity. This improvement highlights the impact of Swin Transformer V2 features in augmenting the network's capacity, crucial for real-world infrared imaging scenarios.

Discussion on loss function. The integration of various loss functions in the SwinIBSR variants markedly influences the perceptual quality of the resulting images. Initially, SwinIBSR-V3, utilizing solely L1 loss, yields smoother images but with limited textural details, as indicated by its higher NIQE scores. The incorporation of perceptual loss in SwinIBSR-V4 enhances textural details, resulting in improved NIQE, PSNR, and SSIM metrics. The addition of adversarial loss in SwinIBSR-V5, while slightly reducing PSNR and SSIM, significantly boosts the visual quality of the images, creating a more realistic and texturally rich visual experience. The inclusion of LDL loss in SwinIBSR further refines the image quality, effectively reducing GAN-related artifacts, which is

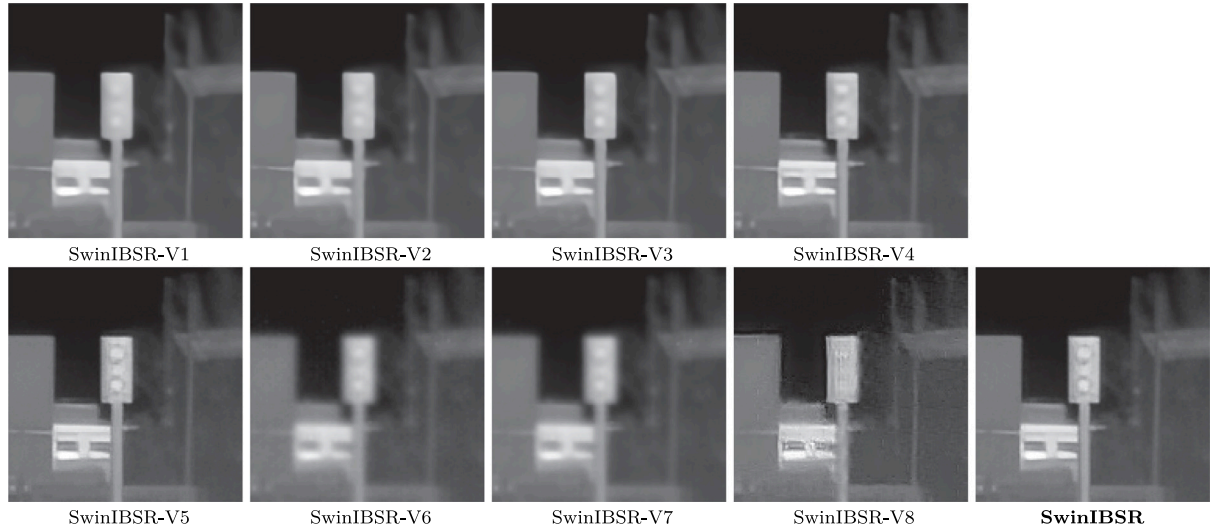


Fig. 5. Visual comparison of SwinIBSR variants.

Table 1

Configuration details of SwinIBSR variants. This table presents the configurations for each variant of the SwinIBSR model, highlighting differences in network structure, loss functions, and training data. The 'Degradation Model' column specifies the degradation model used for each variant. Eq. (5) refers to our proposed real-world infrared image degradation model, designed to accurately simulate the complex degradation processes commonly encountered in real-world infrared imaging scenarios. Other degradation models, indicated by Eq. (3) and Eq. (4), represent simplified degradation processes used for comparative purposes in the ablation study.

Variant	Network structure		Loss function				Training data		Degradation model
	Res-Post-Norm	Scaled-Cosine	$\mathcal{L}_1$	$\mathcal{L}_p$	$\mathcal{L}_{G_s}$	$\mathcal{L}_{LDL}$	Infrared	Visible light	
SwinIBSR-V1			✓				✓	✓	Eq. (5)
SwinIBSR-V2	✓		✓				✓	✓	Eq. (5)
SwinIBSR-V3	✓	✓	✓				✓	✓	Eq. (5)
SwinIBSR-V4	✓	✓	✓	✓			✓	✓	Eq. (5)
SwinIBSR-V5	✓	✓	✓	✓	✓		✓	✓	Eq. (5)
SwinIBSR-V6	✓	✓	✓	✓	✓	✓	✓	✓	Eq. (3)
SwinIBSR-V7	✓	✓	✓	✓	✓	✓	✓	✓	Eq. (4)
SwinIBSR-V8	✓	✓	✓	✓	✓	✓	✓	✓	Eq. (5)
SwinIBSR	✓	✓	✓	✓	✓	✓	✓	✓	Eq. (5)

Table 2

Quantitative comparison of SwinIBSR variants.

Method	Visible light			Infrared
	PSNR↑	SSIM↑	NIQE↓	
SwinIBSR-V1	27.351	0.859	6.411	5.728
SwinIBSR-V2	27.669	0.866	6.385	5.607
SwinIBSR-V3	27.716	0.865	6.302	5.486
SwinIBSR-V4	27.820	0.871	6.019	5.056
SwinIBSR-V5	27.461	0.856	5.743	4.776
SwinIBSR-V6	28.289	0.867	6.268	6.343
SwinIBSR-V7	29.544	0.899	6.127	5.752
SwinIBSR-V8	23.228	0.689	6.530	6.417
SwinIBSR	27.559	0.867	5.193	4.473

evident from the lowest NIQE scores and the enhanced visual details observable in Fig. 5.

Overall, this study demonstrates that while L1 loss provides a stable base for image reconstruction and yields relatively smoother results, the integration of perceptual and adversarial losses, complemented by the LDL loss, play crucial roles in enhancing the perceptual quality and realism of SR images. This combination results in a trade-off, sacrificing high PSNR and SSIM values for visually more striking and detailed outputs.

Discussion on degradation model. The comparison of SwinIBSR variants using different degradation models reveals critical insights. SwinIBSR-V6 and V7, based on simpler degradation equations (Eqs. (3) and (4)), show higher PSNR and SSIM but result in blurrier images, as seen in

Fig. 5. In contrast, the full SwinIBSR model using our advanced real-world degradation model (Eq. (5)) demonstrates superior perceptual quality, reflected in its lower NIQE scores. Despite a slight trade-off in PSNR and SSIM, it more accurately captures complex real-world degradations, producing images that are both visually appealing and realistic. This highlights the necessity of a complex degradation model in infrared image SR for practical, high-quality outcomes.

Discussion on training data. The ablation study of SwinIBSR-V6, which was trained exclusively on infrared data, reveals a significant performance drop, as evidenced by lower PSNR and SSIM values and higher NIQE scores. This decline is further highlighted in Fig. 5, where the results for SwinIBSR-V6 exhibit noticeable noise and artifacts. The likely cause for this degradation in performance is the inherently lower quality of the infrared training data, which may lead to overfitting of the network and reduced ability to generalize effectively to real-world scenarios.

4.3. Comparison of methods

In this section, we conduct a thorough comparison of our proposed SwinIBSR method against previous infrared SR techniques, specifically the methods proposed by Oz et al. [6] and Zou et al. [7]. Additionally, we include SwinIR [44], a prominent method in visible light SR, to broaden the scope of our evaluation. It is important to note that all the compared methods are trained on an identical dataset to ensure fairness in the evaluation. Our comparative analysis encompasses multiple dimensions: objective metrics, visual quality, and the modulation transfer

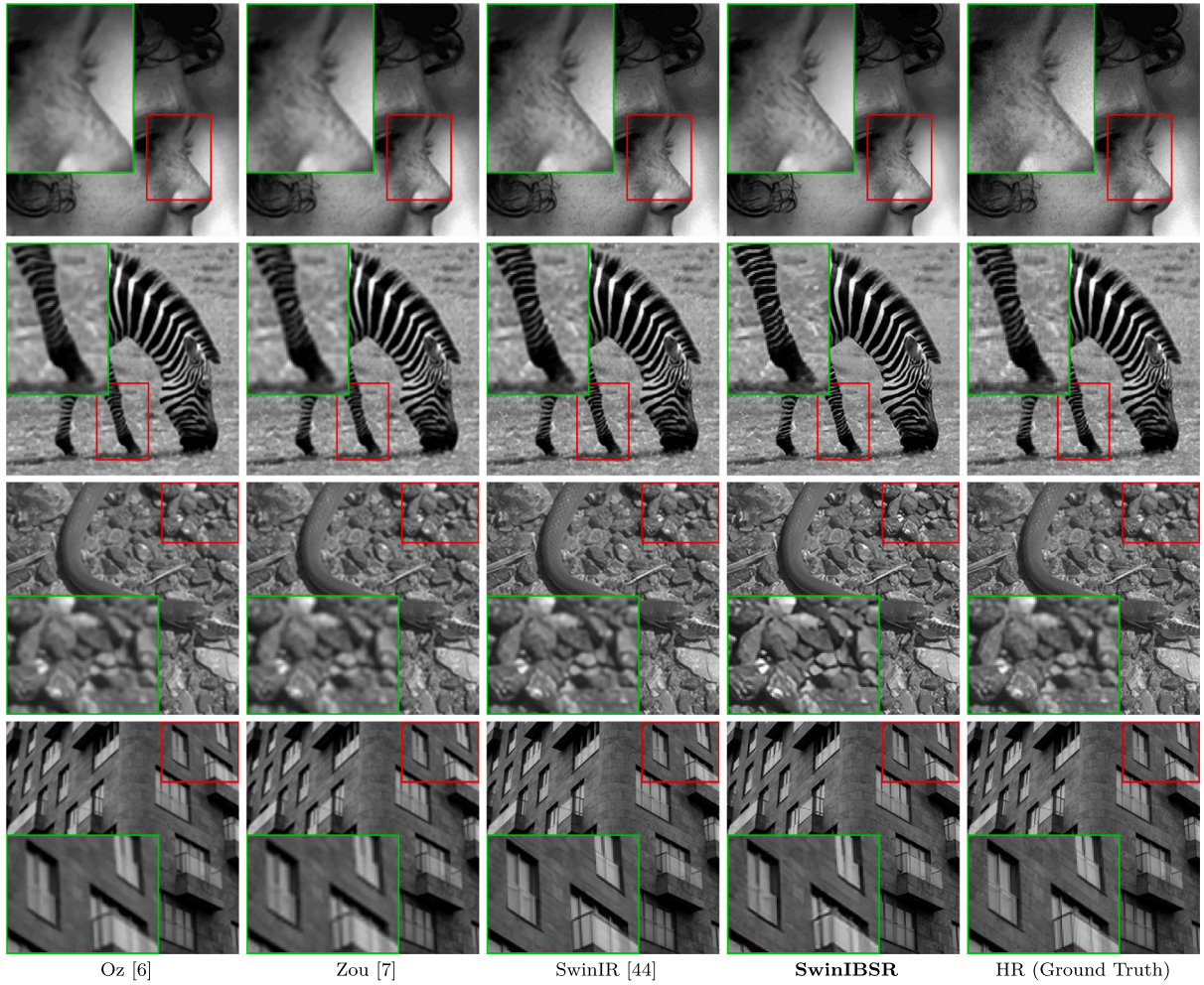


Fig. 6. Visual comparison on visible image SR test datasets. We can see that our method removes noise (first row), suppresses artifacts (second row), and deblurs out-of-focus parts (third row). Zoom in for the best view.

Table 3

Quantitative comparison of various methods on visible light and infrared SR test datasets. Metrics include PSNR, SSIM, and NIQE. The scale factor is $\times 2$.

Method	Parameters	Visible light			Infrared
		PSNR $\uparrow$	SSIM $\uparrow$	NIQE $\downarrow$	NIQE $\downarrow$
Oz [6]	158 K	27.958	0.855	6.172	6.483
Zou [7]	4.6 M	27.238	0.843	5.985	6.116
SwinIR [44]	11.8 M	31.345	0.917	5.487	5.515
SwinIBSR	1.4 M	27.559	0.867	5.193	4.473

function (MTF), providing a multifaceted view of the performance and efficacy of these SR methods.

Objective metrics comparison. Table 3 presents the quantitative results of comparative study. This table includes a comparison of NIQE scores for both visible light and infrared images, alongside the number of parameters for each method. Our method achieves the best NIQE scores in both domains while maintaining a lower parameter count compared to other evaluated methods, highlighting its superior performance and efficiency. Our method achieves the best NIQE scores in both domains, indicative of superior perceptual image quality. This is accomplished while maintaining a lower parameter count, highlighting our method's efficiency.

It is important to reiterate, as also discussed in our 'Ablation Study' section, that while traditional SR evaluation metrics like PSNR and SSIM are included for completeness, they may not always correlate well

with human visual perception. These metrics tend to focus on pixel-level accuracy, which may not fully capture the nuances of perceptual quality evident to the human eye. In line with this understanding, our emphasis on the NIQE score aligns more closely with human visual perception. Therefore, despite the weaker PSNR and SSIM, our method's superior NIQE scores substantiate its capability in generating high visual quality images. This focus on perceptual quality, rather than solely on pixel-level accuracy, is especially pertinent in the context of real-world infrared imaging.

Visual comparison. Since our method is geared towards real-world infrared image SR, visual comparison can provide a more intuitive understanding of its effectiveness. We visually compare SR methods using two types of inputs: LR images from visible image SR test datasets (Fig. 6) and real-world infrared LR images (Fig. 7). These comparisons demonstrate our approach's superior visual quality across both visible light and challenging real-world infrared scenarios. This is attributed to the integration of the real-world degradation model and the use of adversarial loss in our network. These enhancements enable our method to effectively remove blur and noise while simultaneously generating detailed textures. For instance, as evidenced in Fig. 6, our method can even generate SR images that surpass the visual quality of the ground truth, creating more realistic and detailed results.

This capacity for enhancing textures and details introduces noticeable differences at the pixel level compared to the ground truth. Consequently, our method exhibits lower scores in traditional quantitative metrics like PSNR and SSIM, as detailed in Table 3. This trade-off,



Fig. 7. Visual comparison on real-world infrared images. LR ($\times 2$) images are generated by bicubic interpolation. Infrared images are prone to fixed pattern noise due to detector non-uniformity. Although these images undergo preprocessing for non-uniformity correction, some noise remains (second to fifth rows). Additionally, JPEG compression, commonly used to reduce file size, introduces further artifacts (sixth row). Our method demonstrates superiority by producing higher-quality SR images, efficiently reducing noise and artifacts, including the elimination of column striping (second row). Zoom in for the best view.

however, is aligned with our primary objective: to prioritize the creation of visually appealing and realistic images, which is important in real-world applications. The outcomes of visual comparisons further underscore the limitations of traditional metrics like PSNR and SSIM in fully capturing the perceptual quality of images, reaffirming our focus on perceptual image quality as a more relevant measure for real-world applications.

MTF comparison. The modulation transfer function (MTF) [64] is a vital index to quantify infrared imaging systems' imaging quality and spatial resolution. It measures the entire infrared imaging system, including detectors, optical systems, etc. A higher MTF curve indicates better imaging quality. Therefore, the MTF of the SR images is an excellent indicator to evaluate the SR methods for infrared imaging system. We use the slanted-edge method to measure the MTF of the SR images.

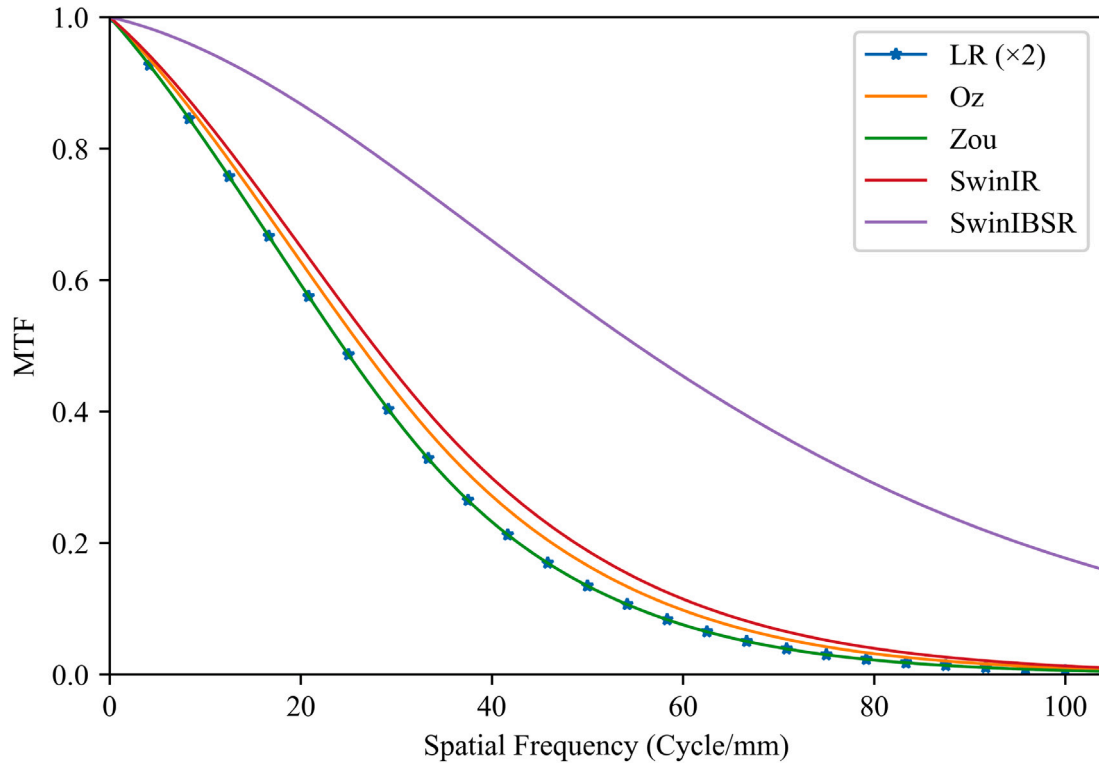
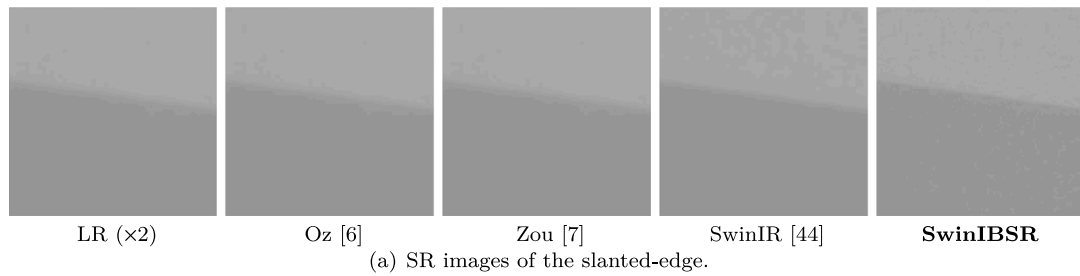


Fig. 8. MTF comparison.

The slanted-edge infrared image is captured by a 640×512 thermal imaging camera with a $15\mu\text{m}$ pixel pitch. Then, the infrared image is processed by SR methods shown in Fig. 8(a). Following the work of [65], the MTF curves are obtained and shown in Fig. 8(b). The MTF curves of SwinIBSR are higher than those of the other methods, which quantitatively indicates that the SR images generated by SwinIBSR have better imaging quality.

5. Conclusion

In this paper, we introduce SwinIBSR, a blind super-resolution (SR) method for real-world infrared images. Our approach centers around a practical degradation model specifically designed to simulate real-world infrared image degradation. Combined with mixed training using infrared and visible light datasets, SwinIBSR significantly improves both the generalization capability of the network and the visual quality of SR images. Extensive experiments demonstrate SwinIBSR's superiority in visual quality over existing methods, validated by objective metrics, visual comparisons, and modulation transfer function analysis. The use of generative adversarial networks in training enables SwinIBSR to produce realistic textures and details, enhancing perceptual image quality.

Future work will explore more efficient network architecture and sophisticated degradation model to further enhance infrared image SR,

pushing the limits of practical applicability and visual fidelity in this challenging domain.

CRediT authorship contribution statement

Yifeng Shi: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation. **Nan Chen:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Yuesheng Pu:** Formal analysis, Conceptualization. **Jiqing Zhang:** Formal analysis, Conceptualization. **Libin Yao:** Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yifeng Shi reports financial support and equipment, drugs, or supplies were provided by Kunming Institute of Physics.

Data availability

The data that has been used is confidential.

References

- [1] Z. Wang, J. Chen, S.C. Hoi, Deep learning for image super-resolution: A survey, *IEEE Trans. Pattern Anal. Mach. Intell.* 43 (10) (2020) 3365–3387.
- [2] J. Li, Z. Pei, T. Zeng, From beginner to master: A survey for deep learning-based single-image super-resolution, 2021, arXiv preprint arXiv:2109.14335.
- [3] C. Sun, J. Lv, J. Li, R. Qiu, A rapid and accurate infrared image super-resolution method based on zoom mechanism, *Infrared Phys. Technol.* 88 (2018) 228–238.
- [4] G. Suryanarayana, E. Tu, J. Yang, Infrared super-resolution imaging using multi-scale saliency and deep wavelet residuals, *Infrared Phys. Technol.* 97 (2019) 177–186.
- [5] T. Yao, Y. Luo, J. Hu, H. Xie, Q. Hu, Infrared image super-resolution via discriminative dictionary and deep residual network, *Infrared Phys. Technol.* 107 (2020) 103314.
- [6] N. Oz, N. Sochen, O. Markovich, Z. Halamish, L. Shpialter-Karol, I. Klapp, Rapid super resolution for infrared imagery, *Opt. Express* 28 (18) (2020) 27196–27209.
- [7] Y. Zou, L. Zhang, C. Liu, B. Wang, Y. Hu, Q. Chen, Super-resolution reconstruction of infrared images based on a convolutional neural network with skip connections, *Opt. Lasers Eng.* 146 (2021) 106717.
- [8] W. Zhang, X. Sui, G. Gu, Q. Chen, H. Cao, Infrared thermal imaging super-resolution via multiscale spatio-temporal feature fusion network, *IEEE Sens. J.* 21 (17) (2021) 19176–19185.
- [9] Y. Huang, Z. Jiang, Q. Wang, Q. Jiang, G. Pang, Infrared image super-resolution via heterogeneous convolutional WGAN, in: *Pacific Rim International Conference on Artificial Intelligence*, Springer, 2021, pp. 461–472.
- [10] A. Hore, D. Ziou, Image quality metrics: PSNR vs. SSIM, in: *2010 20th International Conference on Pattern Recognition*, IEEE, 2010, pp. 2366–2369.
- [11] Q. Huynh-Thu, M. Ghanbari, Scope of validity of PSNR in image/video quality assessment, *Electron. Lett.* 44 (13) (2008) 800–801.
- [12] P. Hanhart, P. Korshunov, T. Ebrahimi, Benchmarking of quality metrics on ultra-high definition video sequences, in: *2013 18th International Conference on Digital Signal Processing, DSP*, IEEE, 2013, pp. 1–8.
- [13] D. Kundu, B.L. Evans, Full-reference visual quality assessment for synthetic images: A subjective study, in: *2015 IEEE International Conference on Image Processing, ICIP*, IEEE, 2015, pp. 2374–2378.
- [14] C. Dong, C.C. Loy, K. He, X. Tang, Image super-resolution using deep convolutional networks, *IEEE Trans. Pattern Anal. Mach. Intell.* 38 (2) (2015) 295–307.
- [15] C. Dong, C.C. Loy, X. Tang, Accelerating the super-resolution convolutional neural network, in: *European Conference on Computer Vision*, Springer, 2016, pp. 391–407.
- [16] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778.
- [17] J. Kim, J.K. Lee, K.M. Lee, Accurate image super-resolution using very deep convolutional networks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 1646–1654.
- [18] Y. Tai, J. Yang, X. Liu, Image super-resolution via deep recursive residual network, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 3147–3155.
- [19] B. Lim, S. Son, H. Kim, S. Nah, K. Mu Lee, Enhanced deep residual networks for single image super-resolution, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2017, pp. 136–144.
- [20] T. Tong, G. Li, X. Liu, Q. Gao, Image super-resolution using dense skip connections, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 4799–4807.
- [21] S. Lei, Z. Shi, X. Wu, B. Pan, X. Xu, H. Hao, Simultaneous super-resolution and segmentation for remote sensing images, in: *IGARSS 2019-2019 IEEE International Geoscience and Remote Sensing Symposium*, IEEE, 2019, pp. 3121–3124.
- [22] S. Lei, Z. Shi, Z. Zou, Super-resolution for remote sensing images via local-global combined network, *IEEE Geosci. Remote Sens. Lett.* 14 (8) (2017) 1243–1247.
- [23] S. Lei, Z. Shi, Hybrid-scale self-similarity exploitation for remote sensing image super-resolution, *IEEE Trans. Geosci. Remote Sens.* 60 (2021) 1–10.
- [24] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, Y. Bengio, Generative adversarial networks, *Commun. ACM* 63 (11) (2020) 139–144.
- [25] M.S. Sajjadi, B. Schölkopf, M. Hirsch, Enhancenet: Single image super-resolution through automated texture synthesis, 2016, arXiv preprint arXiv:1612.07919.
- [26] C. Ledig, L. Theis, F. Huszar, J. Caballero, A. Cunningham, A. Acosta, A. Aitken, A. Tejani, J. Totz, Z. Wang, et al., Photo-realistic single image super-resolution using a generative adversarial network, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 4681–4690.
- [27] X. Wang, K. Yu, C. Dong, C.C. Loy, Recovering realistic texture in image super-resolution by deep spatial feature transform, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 606–615.
- [28] X. Wang, K. Yu, S. Wu, J. Gu, Y. Liu, C. Dong, Y. Qiao, C. Change Loy, Esrgan: Enhanced super-resolution generative adversarial networks, in: *Proceedings of the European Conference on Computer Vision (ECCV) Workshops*, 2018.
- [29] S. Lei, Z. Shi, Z. Zou, Coupled adversarial training for remote sensing image super-resolution, *IEEE Trans. Geosci. Remote Sens.* 58 (5) (2019) 3633–3643.
- [30] J. Johnson, A. Alahi, L. Fei-Fei, Perceptual losses for real-time style transfer and super-resolution, in: *European Conference on Computer Vision*, Springer, 2016, pp. 694–711.
- [31] J. Liang, H. Zeng, L. Zhang, Details or artifacts: A locally discriminative learning approach to realistic image super-resolution, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 5657–5666.
- [32] A. Liu, Y. Liu, J. Gu, Y. Qiao, C. Dong, Blind image super-resolution: A survey and beyond, 2021, arXiv preprint arXiv:2107.03055.
- [33] X. Ji, Y. Cao, Y. Tai, C. Wang, J. Li, F. Huang, Real-world super-resolution via kernel estimation and noise injection, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2020, pp. 466–467.
- [34] K. Zhang, J. Liang, L. Van Gool, R. Timofte, Designing a practical degradation model for deep blind image super-resolution, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 4791–4800.
- [35] X. Wang, L. Xie, C. Dong, Y. Shan, Real-esrgan: Training real-world blind super-resolution with pure synthetic data, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 1905–1914.
- [36] W. Zhang, G. Shi, Y. Liu, C. Dong, X.-M. Wu, A closer look at blind super-resolution: Degradation models, baselines, and performance upper bounds, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 527–536.
- [37] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [38] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, N. Houlsby, An image is worth 16x16 words: Transformers for image recognition at scale, in: *International Conference on Learning Representations*, 2021, URL <https://openreview.net/forum?id=YicbFdNTTy>.
- [39] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, H. Jégou, Training data-efficient image transformers & distillation through attention, in: *International Conference on Machine Learning*, PMLR, 2021, pp. 10347–10357.
- [40] S. Mehta, M. Rastegari, MobileViT: Light-weight, general-purpose, and mobile-friendly vision transformer, in: *International Conference on Learning Representations*, 2022, URL <https://openreview.net/forum?id=vh-0sUt8HIG>.
- [41] H. Chen, Y. Wang, T. Guo, C. Xu, Y. Deng, Z. Liu, S. Ma, C. Xu, C. Xu, W. Gao, Pre-trained image processing transformer, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 12299–12310.
- [42] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, B. Guo, Swin transformer: Hierarchical vision transformer using shifted windows, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 10012–10022.
- [43] Z. Liu, H. Hu, Y. Lin, Z. Yao, Z. Xie, Y. Wei, J. Ning, Y. Cao, Z. Zhang, L. Dong, et al., Swin transformer v2: Scaling up capacity and resolution, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 12009–12019.
- [44] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, R. Timofte, Swinir: Image restoration using swin transformer, in: *2021 IEEE/CVF International Conference on Computer Vision Workshops, ICCVW*, 2021, pp. 1833–1844, <http://dx.doi.org/10.1109/ICCVW54120.2021.00210>.
- [45] F. Yang, H. Yang, J. Fu, H. Lu, B. Guo, Learning texture transformer network for image super-resolution, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 5791–5800.
- [46] Z. Lu, J. Li, H. Liu, C. Huang, L. Zhang, T. Zeng, Transformer for single image super-resolution, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 457–466.
- [47] S. Lei, Z. Shi, W. Mo, Transformer-based multistage enhancement for remote sensing image super-resolution, *IEEE Trans. Geosci. Remote Sens.* 60 (2021) 1–11.
- [48] Y.-Q. Liu, X. Du, H.-L. Shen, S.-J. Chen, Estimating generalized Gaussian blur kernels for out-of-focus image deblurring, *IEEE. Trans. Circ. Syst. Video Technol.* 31 (3) (2020) 829–843.
- [49] Z. He, Y. Cao, Y. Dong, J. Yang, Y. Cao, C.-L. Tisse, Single-image-based nonuniformity correction of uncooled long-wave infrared detectors: A deep-learning approach, *Appl. Opt.* 57 (18) (2018) D155–D164.
- [50] G.K. Wallace, The JPEG still picture compression standard, *Commun. ACM* 34 (4) (1991) 30–44.
- [51] T. Xiao, M. Singh, E. Mintun, T. Darrell, P. Dollár, R. Girshick, Early convolutions help transformers see better, *Adv. Neural Inf. Process. Syst.* 34 (2021) 30392–30400.
- [52] W.-S. Lai, J.-B. Huang, N. Ahuja, M.-H. Yang, Fast and accurate image super-resolution with deep laplacian pyramid networks, *IEEE Trans. Pattern Anal. Mach. Intell.* 41 (11) (2018) 2599–2613.
- [53] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, 2014, arXiv preprint arXiv:1409.1556.
- [54] E. Agustsson, R. Timofte, Ntire 2017 challenge on single image super-resolution: Dataset and study, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2017, pp. 126–135.

- [55] R. Timofte, E. Agustsson, L. Van Gool, M.-H. Yang, L. Zhang, Ntire 2017 challenge on single image super-resolution: Methods and results, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2017, pp. 114–125.
- [56] I. Loshchilov, F. Hutter, Decoupled weight decay regularization, in: *International Conference on Learning Representations*, 2019, URL <https://openreview.net/forum?id=Bkg6RiCqY7>.
- [57] D.P. Kingma, J. Ba, Adam: A method for stochastic optimization, in: *International Conference on Learning Representations*, 2015.
- [58] I. Loshchilov, F. Hutter, SGDR: Stochastic gradient descent with warm restarts, in: *International Conference on Learning Representations*, 2017, URL <https://openreview.net/forum?id=Skq89Scxx>.
- [59] M. Bevilacqua, A. Roumy, C. Guillemot, M. line Alberi Morel, Low-complexity single-image super-resolution based on nonnegative neighbor embedding, in: *Proceedings of the British Machine Vision Conference*, BMVA Press, 2012, pp. 135.1–135.10, <http://dx.doi.org/10.5244/C.26.135>.
- [60] R. Zeyde, M. Elad, M. Protter, On single image scale-up using sparse-representations, in: J.-D. Boissonnat, P. Chenin, A. Cohen, C. Gout, T. Lyche, M.-L. Mazure, L. Schumaker (Eds.), *Curves and Surfaces*, Springer Berlin Heidelberg, Berlin, Heidelberg, ISBN: 978-3-642-27413-8, 2012, pp. 711–730.
- [61] D. Martin, C. Fowlkes, D. Tal, J. Malik, A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics, in: *Proceedings Eighth IEEE International Conference on Computer Vision. ICCV 2001*, Vol. 2, 2001, pp. 416–423, <http://dx.doi.org/10.1109/ICCV.2001.937655>.
- [62] J.-B. Huang, A. Singh, N. Ahuja, Single image super-resolution from transformed self-exemplars, in: *2015 IEEE Conference on Computer Vision and Pattern Recognition, CVPR*, 2015, pp. 5197–5206, <http://dx.doi.org/10.1109/CVPR.2015.7299156>.
- [63] A. Mittal, R. Soundararajan, A.C. Bovik, Making a “completely blind” image quality analyzer, *IEEE Signal Process. Lett.* 20 (3) (2012) 209–212.
- [64] J.W. Goodman, P. Sutton, Introduction to Fourier optics, *Quant. Semiclassical Opt.-J. Eur. Opt. Soc. B* 8 (5) (1996) 1095.
- [65] S. Najafi, K. Madanipour, Measurement of the modulation transfer function of a charge-coupled device array by the combination of the self-imaging effect and slanted edge method, *Appl. Opt.* 52 (19) (2013) 4724–4727.